

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 1 Part A and B— Uninformed Search Algorithms

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GitHub Repository Link:

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1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	Yes
Estimated percentage of the work influenced by AI	11%
Did you attach evidence of AI use where applicable?	Yes

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Colab(Gemini)	When the code is not running so to help debug errors when my code didn't run	There is a button to click(explain the error) so understand "Why is my expand method failing?" "What does this error mean?" "Explain the NotImplementedError"	- Debugging the expand method - Fixing GridProblem.is_goal - Understanding error messages	I read the error messages first, then used AI to explain them. I wrote the fixes myself and tested the code by running it.
Gemini (Google Colab AI)	To understand error messages and fix code	To understand "What does AssertionError mean?" "Why is Greedy failing?"	Fixing algorithm test failures, debugging Greedy Best-First Search	I read errors, asked AI for explanation, fixed code myself, and retested

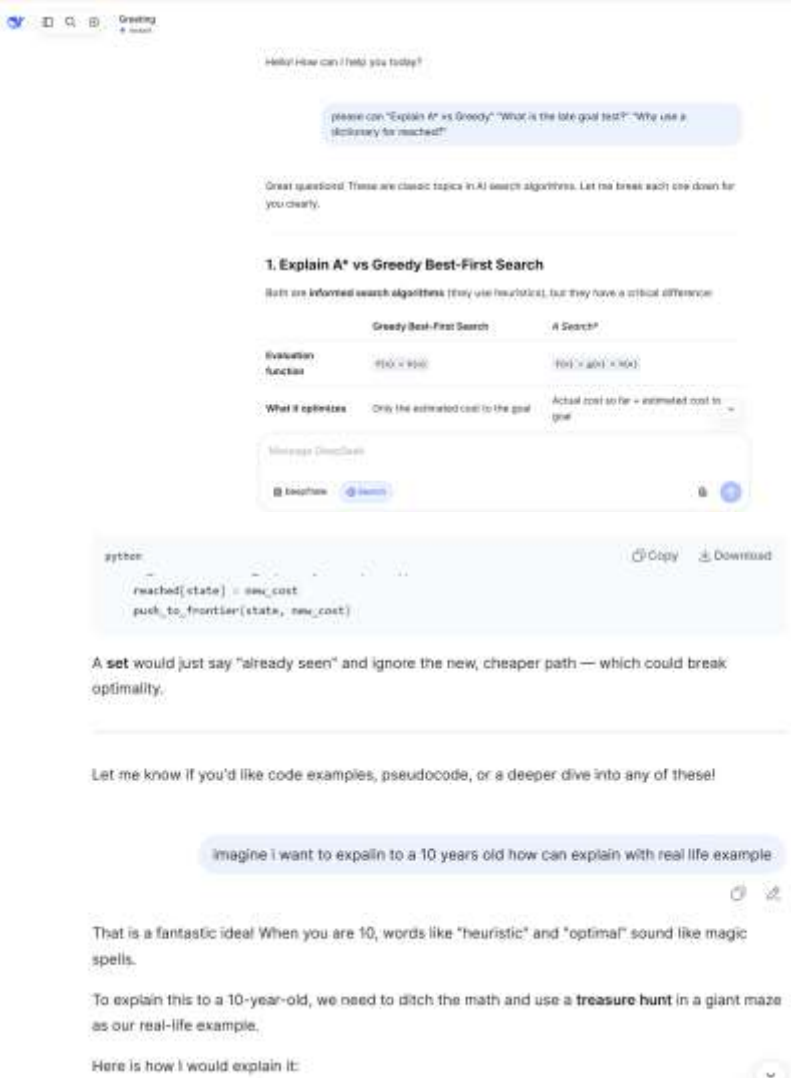
Deepseek	To understand search algorithm concepts	"Explain A* vs Greedy" "What is the late goal test?" "Why use a dictionary for reached?"	- Understanding heuristic functions - Understanding the BestFirstSearch framework - Completing reflection questions	I read the explanations, then wrote the answers in my own words. I tested the algorithms on sample maps to confirm they work correctly.
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3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft	No	
Prompt history	No	

	Screenshot of AI interaction Yes The top part of the screenshot displays a Jupyter Notebook editor with Python code implementing search algorithms like Uniform Cost Search and Greedy Search. Below the notebook are two terminal windows. <pre>*** NotImplementedError Traceback (most recent call last) /tmp/ipynkernel_3198/2689612982.py in <cell line: 0>() 12 UniformCostSearch(), 13 > --> 14 res = algo.search(check_problem) 15 print(f"[algo.algorithm_name] returned {res.status}") 16 assert res.status == "success", f"{res.algorithm} failed to find a path"</pre> <pre>... frames ... /tmp/ipynkernel_3198/1305071832.py in expand(self, problem, node) 113 # cost <- node.PATH_COST + problem.ACTION_COST(s, action, s_prime) 114 # yield NODE(STATE=s_prime, PARENT=node, ACTION=action, PATH_COST=cost) --> 115 raise NotImplementedError("Complete SearchAlgorithm.expand") 116 @abstractmethod 117 def search(self, problem: Problem) -> SearchResult:</pre> NotImplementedError: Complete SearchAlgorithm.expand Next steps: Explain error <pre>print("Algorithm self-check passed.")</pre> <pre>*** Greedy returned failure AssertionError Traceback (most recent call last) /tmp/ipynkernel_3190/2689612982.py in <cell line: 0>() 14 res = algo.search(check_problem) 15 print(f"[algo.algorithm_name] returned {res.status}") --> 16 assert res.status == "success", f"{res.algorithm} failed to find a path" 17 assert res.resolution_cost == 4, f"{res.algorithm} returned cost {res.resolution_cost}" 18 assert res.path[0] == (0, 0) and res.path[-1] == (2, 2)</pre> AssertionError: Greedy failed to find a path Next steps: Explain error <tr> <td data-bbox="181 1629 339 1848">Exported AI conversation</td><td data-bbox="339 1629 1424 1848"></td></tr> <tr> <td data-bbox="181 1848 339 1848">AI-generate</td><td data-bbox="339 1848 1424 1848"></td></tr>	Exported AI conversation		AI-generate	
Exported AI conversation					
AI-generate					

d code snippet											
Revised version showing student input	Yes	 The screenshot shows a chat interface with a user asking for an explanation of A* vs Greedy Best-First Search. The AI assistant responds with a detailed comparison table and code snippets. The table compares Greedy Best-First Search (f(x) = g(x)) and A* Search (f(x) = g(x) + h(x)). The AI also provides a code snippet for a search algorithm and explains the importance of a set to track visited states. The user then asks for a real-life example, and the AI explains it using a treasure hunt in a maze. <table border="1"><thead><tr><th></th><th>Greedy Best-First Search</th><th>A* Search</th></tr></thead><tbody><tr><td>Evaluation function</td><td>$f(x) = g(x)$</td><td>$f(x) = g(x) + h(x)$</td></tr><tr><td>What it optimizes</td><td>Only the estimated cost to the goal</td><td>Actual cost so far + estimated cost to goal</td></tr></tbody></table> <pre>python def reached(state): return state == goal def push_to_frontier(state, new_cost): frontier.append((state, new_cost))</pre> A set would just say "already seen" and ignore the new, cheaper path — which could break optimality. Let me know if you'd like code examples, pseudocode, or a deeper dive into any of these! imagine i want to explain to a 10 years old how can explain with real life example That is a fantastic idea! When you are 10, words like "heuristic" and "optimal" sound like magic spells. To explain this to a 10-year-old, we need to ditch the math and use a treasure hunt in a giant maze as our real-life example. Here is how I would explain it:		Greedy Best-First Search	A* Search	Evaluation function	$f(x) = g(x)$	$f(x) = g(x) + h(x)$	What it optimizes	Only the estimated cost to the goal	Actual cost so far + estimated cost to goal
	Greedy Best-First Search	A* Search									
Evaluation function	$f(x) = g(x)$	$f(x) = g(x) + h(x)$									
What it optimizes	Only the estimated cost to the goal	Actual cost so far + estimated cost to goal									

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes

My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Student Signature: _____ Ramatou AMIDOU _____

Date: _____ 20/06/2026 _____